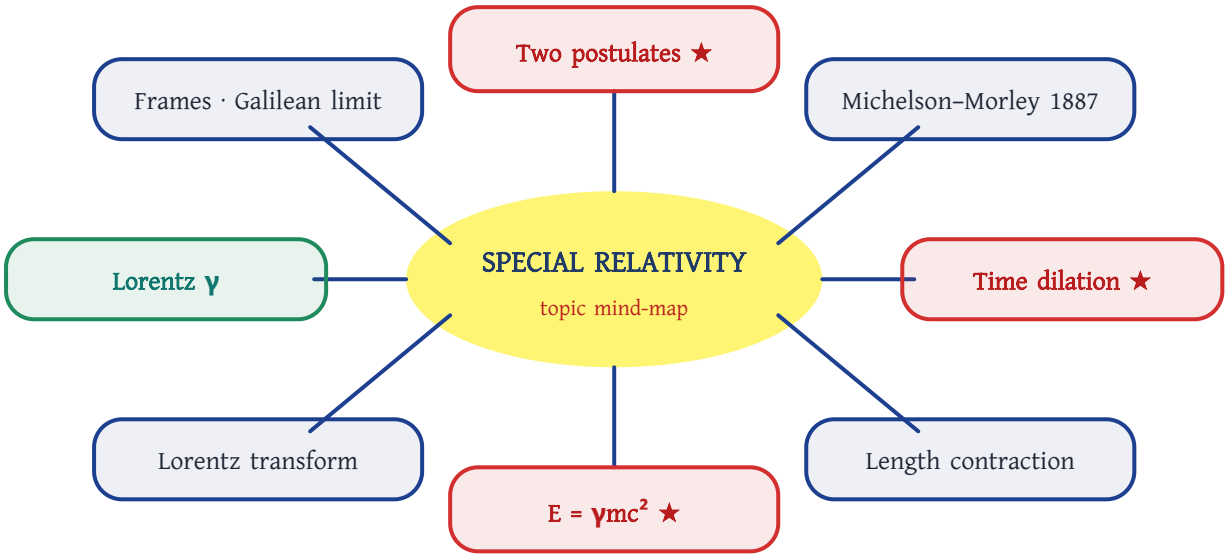
# Special Relativity

Postulates · Lorentz γ · $\Delta t = \gamma \Delta t_0$ · $L = L_0/\gamma$ · $E = \gamma mc^2$: exam-ready in 11 sheets




↑ the whole topic on one page: red = high-yield, green = the one number everything hangs on

## Syllabus coverage

Basic Physics lists seven lines; the third is "**theory of special relativity**". This notebook covers exactly that line. Photons and the quantum theory of light belong to the Quantum Physics lesson.

**High-yield list**

① $c = 3 \times 10^8$ m/s, same in all frames
② $\gamma = 1/\sqrt{(1 - v^2/c^2)}$, $\gamma \geq 1$
③ $\Delta t = \gamma \Delta t_0$ · $L = L_0/\gamma$
④ $u = (u' + v)/(1 + u'v/c^2)$
⑤ $E = \gamma mc^2$ · $K = (\gamma - 1)mc^2$
⑥ $E^2 = p^2c^2 + m_0^2c^4$
⑦ $L_0$, $\Delta t_0$ = proper length, proper time

**Legend:** **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

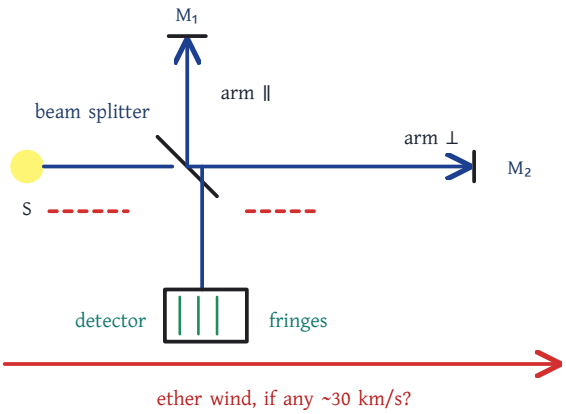
## Contents


✓ Frames, ether & Michelson–Morley … p. 2
✓ Einstein's two postulates (1905) … p. 3
✓ The Lorentz factor γ … p. 4
✓ Lorentz transformation … p. 5
✓ Velocity addition … p. 5
✓ Time dilation $\Delta t = \gamma \Delta t_0$ … p. 6
✓ Length contraction $L = L_0/\gamma$ … p. 7
✓ Simultaneity & the light cone … p. 8
✓ Mass, energy & momentum … p. 9
✓ Night-before p. 10 · Practice p. 11

# 1 Frames, the Ether & Michelson–Morley

**Definition:** An **inertial frame** is one in which Newton's first law holds — a free body moves in a straight line at constant speed. Special relativity compares **only inertial frames**. Classical mechanics assumes **absolute time and length**: $t' = t$, $x' = x - vt$, correct only while $v \ll c$.

> Newton: "time flows the same everywhere"




equal perpendicular arms: only an "ether wind" could shift the fringes

**Michelson–Morley, 1887.** At the Case School of Applied Science, Cleveland. A beam is split by a **beam splitter** into two **equal, perpendicular** arms; mirrors return them to recombine. A stationary luminous ether should make the outward and return legs unequal, so rotating the apparatus should drag the fringe pattern.

| Quantity | Value |
|---|---|
| Predicted fringe shift | ≈ **0.4** fringe |
| Shift actually observed | < **0.01** fringe |
| Ether drift still allowed | a few km/s at most |

**Memory trick:** The arms are **perpendicular and equal**, so in a real ether only the **round trip** differs — "parallel arm gains, perpendicular arm is dragged". The predicted shift is a **fraction of one fringe**, which is exactly why the test is so delicate.

**Exam trap:** A **null result removes a hypothesis; it does not prove another**. Michelson won the **1907** Nobel Prize in Physics — the first American to do so — for precision optical instruments, **not** for relativity.

**Worked example ✍️ how big was the predicted shift?:** For an arm of length $L$ and an ether drift $v$, $t_{\parallel} = L/(c-v) + L/(c+v) = 2Lc/(c^2-v^2)$, $t_{\perp} = 2L/\sqrt{(c^2-v^2)}$. With $v = 30$ km/s, $v/c = 10^{-4}$, so $1/(1-\beta^2) \approx 1 + \beta^2$ and $1/\sqrt{(1-\beta^2)} \approx 1 + \beta^2/2$: $\Delta t \approx (2L/c)(\beta^2/2) = L\beta^2/c$. For $L = 11$ m and $\lambda = 550$ nm, $N = c\Delta t/\lambda \approx$ **0.2 fringe**, the classical prediction of the order of 0.4 fringe — and the experiment saw less than a hundredth of it.

**Still the same technique:** Splitting one beam into two paths and comparing them is how LIGO detects gravitational waves and how stepper lithography prints features a few nanometres wide. The null result ended the ether; the interferometer never went out of fashion.

## 2 Einstein's Two Postulates (1905)

**Why it was needed:** Maxwell's equations give one speed for light in vacuum and no other. Galilean kinematics plus an ether predicted the same laws everywhere **except** in the ether frame — so ether physics would be the one exception. Einstein's 1905 paper *On the Electrodynamics of Moving Bodies* removed the exception instead of explaining it.

**The two postulates:** ① **Principle of relativity** — the laws of physics are **identical in every inertial frame**; no closed-laboratory experiment can detect uniform absolute motion. ② **Constancy of the speed of light** — light in vacuum travels at $c = 3 \times 10^8$ **m/s** in *every* inertial frame, whatever the motion of the source or of the observer.

| Postulate | Rules out | Forces | Frequent MCQ hook |
|---|---|---|---|
| ① Relativity principle | absolute rest, a privileged ether frame | all inertial frames are physically equivalent | Galilean mechanics is the $v \ll c$ limit |
| ② c invariant | the ether drift of Michelson–Morley | no absolute time; $\gamma$ finite only for $v < c$ | c is exact in SI since 1983 |

| Class | Galileo / Newton (1642–1727) | Einstein / Lorentz (1905) |
|---|---|---|
| Time | absolute: $t' = t$ | mixed with space: $t' = \gamma(t - vx/c^2)$ |
| Length | absolute: $x' = x - vt$ | mixed with time: $x' = \gamma(x - vt)$ |
| Velocity sum | $u = u' + v$ | $u = (u' + v)/(1 + u'v/c^2)$ |
| Speed limit | none | c |

**Worked example ✍️:** A pulse leaves a stationary source. Postulate 2 says every observer measures c. Postulate 1 says the *laws* hold in all frames, so the same equations that give the usual Doppler shift in air must give the **relativistic Doppler shift** in vacuum: for an observer receding at v, $f' = f\sqrt{((1-\beta)/(1+\beta))}$.

**Memory trick:** "The **physics** is the same, **light** is the same." Postulate 1: all observers agree on the laws. Postulate 2: all observers agree on *one* speed — and that one speed is what mixes space with time.

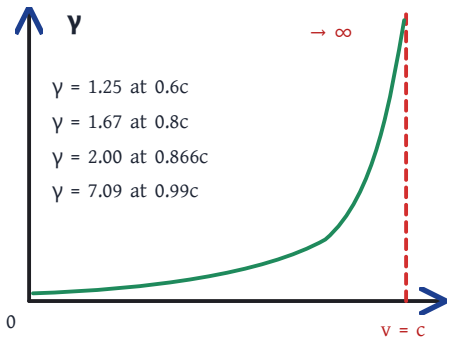
**Common misconception:** c is not "the speed of light" in general — it is the speed **in vacuum**, and since **1983** the metre is defined so that **c = 299 792 458 m/s exactly**. In glass, water or air the speed is **c/n**.

**One line on the general theory:** The **general theory (1915)** drops the inertial-frame restriction and treats gravitation as curvature of spacetime. That is a separate lesson: here every frame is inertial and gravity is not in the equations at all.

## 3 The Lorentz Factor γ

**Definition:** Every formula in special relativity carries one number, the **Lorentz factor**: $\boldsymbol{\gamma = 1/\sqrt{(1 - v^2/c^2)} = 1/\sqrt{(1 - \beta^2)}}$, with $\beta = v/c$. Properties: $\boldsymbol{\gamma \geq 1}$ always, $\gamma = 1$ at $v = 0$, $\gamma \to \infty$ as $v \to c$, and $\gamma \approx 1 + \beta^2/2$ when $v \ll c$ (the Galilean limit).

| v/c | $\beta^2$ | $\sqrt{(1-\beta^2)}$ | γ | 1/γ |
|---|---|---|---|---|
| 0 | 0 | 1.000 | **1.000** | 1.000 |
| 0.20 | 0.04 | 0.980 | 1.021 | 0.980 |
| 0.40 | 0.16 | 0.917 | 1.091 | 0.917 |
| 0.50 | 0.25 | 0.866 | 1.155 | 0.866 |
| **0.60** | 0.36 | 0.800 | **1.250** | 0.800 |
| **0.80** | 0.64 | 0.600 | **1.667** | 0.600 |
| 0.866 = √3/2 | 0.75 | 0.500 | **2.000** | 0.500 |
| 0.90 | 0.81 | 0.436 | 2.294 | 0.436 |
| 0.99 | 0.9801 | 0.141 | **7.089** | 0.141 |




flat near $v = 0$, and $\propto \infty$ at $v = c$

**Worked example 1 ✍️:** γ at 0.6c: $1/\sqrt{0.64} = 1/0.8 = \mathbf{1.25}$ · at 0.8c: $1/\sqrt{0.36} = 1/0.6 = \mathbf{5/3 = 1.67}$ · at 0.99c: $1/\sqrt{0.0199} = \mathbf{7.09}$. LHC protons run at $\gamma \approx 7\,500$.

**Memory trick:** γ is the number you **multiply time by** and the number you **divide length by**. $\sqrt{(1 - v^2/c^2)}$ is just **1/γ** — never γ.

**Worked example 2: the inverse ✍️:** "What speed gives $\gamma = 2$?" From $\gamma^2(1 - v^2/c^2) = 1 \to v = c\sqrt{(1 - 1/\gamma^2)}$.
$\gamma = 2 \to v = c\sqrt{(3/4)} = \mathbf{0.866c}$. Check the other way: $\gamma = 1.25 \to v = c\sqrt{(1 - 1/1.5625)} = c\sqrt{0.36} = \mathbf{0.60c}$ ✓

**Exam trap:** (i) $\gamma(0.6c) = \mathbf{1.25}$, not 1.67 — 1.67 belongs to 0.8c. (ii) γ can never be **less than 1**; if your answer is 0.8 you have computed $\sqrt{(1 - \beta^2)}$. (iii) $\gamma \to \infty$ does **not** forbid c: it forbids a body with mass *reaching* c.

**All the results in one line each:**

**Time:** $\Delta t = \gamma \Delta t_0$

**Length:** $L = L_0/\gamma$

**Mass:** $m = \gamma m_0$

**Energy:** $E = \gamma m_0 c^2$

**Momentum:** $p = \gamma m_0 v$

**Velocity sum:** $u = (u'+v)/(1+u'v/c^2)$

**IT link:** A Lorentz boost is a 2 × 2 matrix on (x, ct) — the form lattice gauge-theory codes and relativistic engines actually use:

```
gamma = 1 / sqrt(1 - v*v/(c*c));
Lambda = [[ gamma,     -gamma*v/c ],
          [ -gamma*v/c,  gamma ]];  /* boost on (x, ct) */
```

## 4 The Lorentz Transformation

**Definition:** The rules that take an event from the **rest frame S′** of an object to the lab frame **S** in which the object moves along x at speed v:

**The four equations:** $x' = \gamma(x - vt)$ · $t' = \gamma(t - vx/c^2)$ · $y' = y$ · $z' = z$

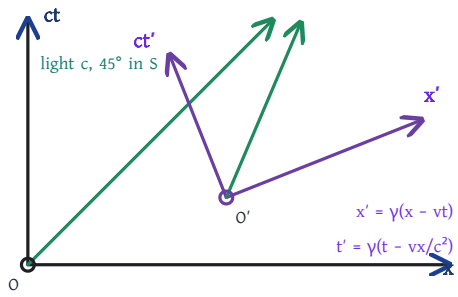
Inverse (swap the primes, flip the sign of v): $x = \gamma(x' + vt')$ · $t = \gamma(t' + vx'/c^2)$

**Derivation sketch ✍️ why the coefficients must be γ:**

1. Space and time are homogeneous, so the transform is **linear**: $x' = a(x - vt)$, $t' = b(t - k \cdot v \cdot x)$
2. Postulate 2 for the ray x = ct, which must also have x′ = ct′: $a(c-v) = b(c - kvc) \Rightarrow$ **a = b** and $\mathbf{k = 1/c^2}$
3. The transform for −v must be the **inverse** of the one for +v; with a = b = A this gives $A^2(1 - v^2/c^2) = 1$, so **A = γ**

The **interval** $c^2t^2 - x^2 - y^2 - z^2$ is the same in every inertial frame, and so is c. That is the whole reason time and length cannot both stay absolute.

| Frame-dependent | Form | Frame-invariant |
|---|---|---|
| Time, length, momentum | $\gamma\Delta t_0 \cdot L_0/\gamma \cdot \gamma m_0 v$ | **c**, the interval $c^2t^2 - x^2 - y^2 - z^2$, $\mathbf{m_0}$, $\mathbf{m_0c^2}$, **charge**, and the causal order of timelike events |
| Energy | $\gamma m_0 c^2$ | |




the same 45° light ray in both frames — that is what fixes γ

## 5 Relativistic Velocity Addition

**Composition:** If an object moves at u′ in S′ and S′ moves at v in S, then $\mathbf{u = (u' + v)/(1 + u'v/c^2)}$. Putting u′ = c returns u = c exactly, so **c is the ceiling of every speed.**

**Worked example 1 ✍️:** Two ships head the same way, each seeing the other at 0.6c. Speed of one relative to Earth: u = (0.6c + 0.6c)/(1 + 0.36) = **0.882c** — **not** 1.2c.

**Worked example 2 ✍️:** Two bodies recede at 0.8c and 0.6c relative to Earth: u = (0.8c + 0.6c)/(1 + 0.8 × 0.6) = **0.946c**.

**Memory trick:** "γ in, γ out": every relativistic formula is the Newtonian one times γ (time, momentum, energy) or divided by γ (length).

**Exam trap:** Galilean addition $u = u' + v$ is **wrong** except in the $v \ll c$ limit. $0.6c + 0.6c = 0.882c$, never $1.2c$.

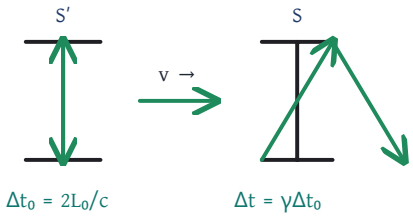
## 6 Time Dilation $\Delta t' = \gamma \Delta t$

**Definition:** Two events at the **same place in one inertial frame** are separated there by the **proper time** $\Delta t_0$ — the interval read by one clock present at both. Every other inertial frame measures a longer interval: $\Delta t' = \gamma \Delta t_0 = \Delta t_0 / \sqrt{(1 - v^2/c^2)}$, and always $\Delta t' \geq \Delta t_0$.

> books write it $\Delta t = \gamma \Delta t_0$ or $\Delta t' = \gamma \Delta t$ — check which symbol is the proper time

**Derivation A: the light clock ✍️:** A pulse bounces between two mirrors a proper distance $L_0$ apart, in a clock moving at $v$.

1. Own frame S′: straight up and down, so $\Delta t_0 = 2L_0/c$
2. Lab S: each leg is the hypotenuse of a right triangle with legs $L_0$ and $v\Delta t/2$
3. $c\Delta t/2 = \sqrt{(L_0^2 + v^2\Delta t^2/4)} \implies \Delta t^2(c^2 - v^2) = 4L_0^2 \implies$ **$\Delta t = \gamma \Delta t_0$**




each leg: hypotenuse $L_0\sqrt{2}$

the same two ticks; the tilted path takes $\sqrt{2}$ longer

**Derivation B: from the Lorentz transform ✍️:** Both ticks are at the same place in S′, so $\Delta x' = 0$.

1. $\Delta x' = \gamma(\Delta x + v\Delta t') = 0 \implies \Delta x = -v\Delta t'$
2. $\Delta t' = \gamma(\Delta t - v\Delta x/c^2) = \Delta t/\gamma \implies$ **$\Delta t = \gamma \Delta t'$**

**Worked example 1: the muon ✍️:** Proper lifetime $\tau_0 = 2.2$ μs at $v = 0.8c$, $\gamma = 5/3$. $\Delta t = (5/3)(2.2\ \mu s) =$ **3.67 μs**; the lab path is $v\Delta t = 0.8c \times 3.67\ \mu s =$ **880 m**.

**Worked example 2: cosmic rays ✍️:** At $v = 0.998c$, $\gamma = 1/\sqrt{0.004} =$ **15.8**, so $\Delta t = 15.8 \times 2.2$ μs = **34.8 μs** and the muon covers ≈ **10.4 km** — muons born high in the atmosphere reach the ground.

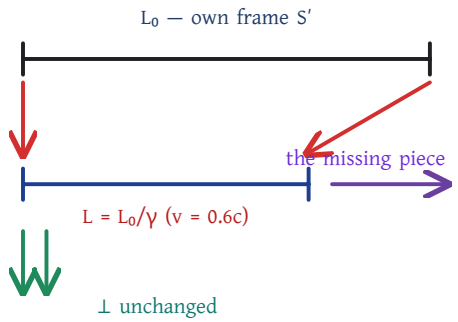
**Memory trick:** "**Proper time is the shortest time.**" One clock in one place reads the smallest interval; every other frame over-measures it, because $\gamma \geq 1$.

**Exam trap:** Both observers see the other's clock run slow — no contradiction, they are in **different** inertial frames. Only when one twin **turns round** do the readings differ, and then the stay-at-home twin is **older**.

**Applications & IT link:** **Hafele–Keating (1971):** four caesium clocks flown round the world on jets came back shifted by exactly the amount γ and the difference in gravitational potential predict. **GPS:** satellite clocks gain ≈ **38 μs per day** overall.

## 7 Length Contraction L′ = L/γ

**Definition:** A body moving along the line joining two observers has its length measured **smaller** than its **proper length**: $L = L_0\sqrt{(1 - v^2/c^2)} = L_0/\gamma$, where $L_0$ is the length in the body's own rest frame. Only the component **parallel to the motion** contracts; every transverse dimension is unchanged.




the shortened rod is read with **both ends at one instant of S**

**Derivation ✍ (one line):** The rod is at rest in S′, so its ends differ by $\Delta x' = L_0$. In S both end positions must be recorded at the **same instant**, $\Delta t = 0$, so the time term vanishes and $x'_2 - x'_1 = \gamma(x_2 - x_1) = \gamma L$. Hence $L_0 = \gamma L \Longrightarrow L = L_0/\gamma$.

**Worked example ✍:** A spaceship has proper length $L_0$ = 120 m and passes Earth at 0.8c (γ = 5/3): L = 120 × 0.6 = **72 m** in the Earth frame, while its height and width are unchanged — 72 m long, H high, W wide.

| v | 0.20c | 0.50c | 0.60c | 0.80c | 0.866c | 0.90c | 0.99c |
|---|---|---|---|---|---|---|---|
| **γ** | 1.021 | 1.155 | 1.25 | 1.667 | 2.000 | 2.294 | 7.089 |
| **$L/L_0 = \sqrt{(1-\beta^2)}$** | 0.980 | 0.866 | 0.800 | 0.600 | 0.500 | 0.436 | 0.141 |

**Memory trick:** "γ divides length, γ multiplies time, and it only touches the **along-motion** direction." Read 1/γ straight off the γ table: at 0.8c it is 0.6, so the rod keeps 60 % of its length.

**Exam trap:** (i) Contraction is **not** an optical illusion of light arriving late — it survives unlimited resolution. (ii) It is **not** perpendicular to the motion. (iii) The **proper** length $L_0$ is what the object itself measures. (iv) Only **inertial** observers can quote $L = L_0/\gamma$.

**Worked example ✍ the LHC ring:** The Large Hadron Collider ring is 27.4 km long in the laboratory frame. The protons circling it at γ ≈ 7 500 see their own orbit as only 27.4 km / 7 500 ≈ **3.7 m** of proper circumference. That is the "length-contracted universe" that makes particle physics convenient, and it is the same equation turned round.

**Reading the sky:** Astrophysicists invert the same formula: a strongly redshifted galaxy is receding close to c, and its surface brightness, angular size and light-travel time must all be corrected with these factors. A quasar at z = 6 has been travelling since the Universe was about 1 Gyr old.

## 8 Relativity of Simultaneity, Spacetime & the Light Cone

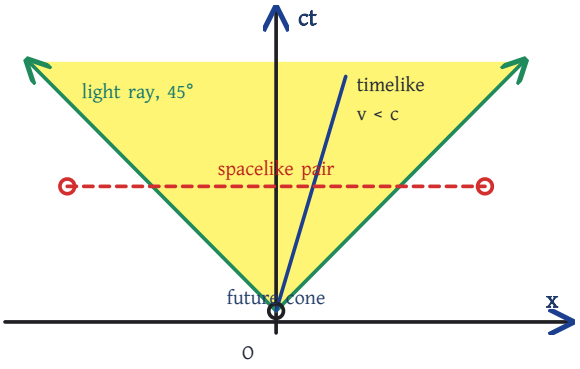
**Definition:** Two events simultaneous in one frame ($\Delta t = 0$) are **not** simultaneous in a frame moving along the line joining them: $\Delta t' = -\gamma v \Delta x / c^2 \neq 0$. Their order is therefore relative. Spacetime is the single 4-dimensional entity (x, y, z, t); Minkowski (1908) showed its geometry is not Euclidean.

| Separation | $\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$ | All frames must agree that |
|---|---|---|
| **Timelike** ($\Delta x < c\Delta t$) | $> 0$ | the cause comes first |
| **Null / lightlike** ($\Delta x = c\Delta t$) | $= 0$ | only light does this |
| **Spacelike** ($\Delta x > c\Delta t$) | $< 0$ | order **can** differ; no signal connects them |

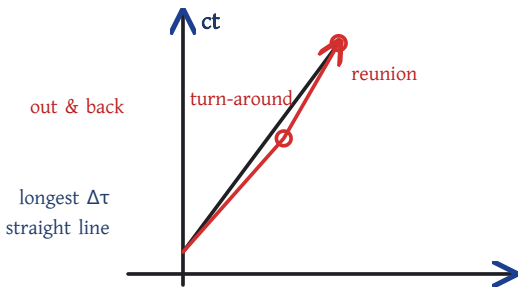
**Worked example ✍️:** Two flashes occur 1 000 km apart along a rail. An observer S at the front records them **simultaneously**, so $\Delta t = 0$, $\Delta x = 1\,000$ km. A train passes at $v = 0.6c$ ($\gamma = 1.25$). In the train frame $\Delta t' = -\gamma v \Delta x / c^2 = -(1.25)(0.6c)(1.0 \times 10^6\ \text{m})/c^2 = -0.75 \times 10^6/(3 \times 10^8)\ \text{s} =$ **−2.5 ms**: the rear flash happens 2.5 ms **earlier** in the train frame.

**Memory trick:** "Set **$\Delta t = 0$** in the transform and simultaneity splits." A pair of events one frame calls simultaneous gains a gap $\gamma v \Delta x / c^2$ in the other frame.

**Exam trap:** Simultaneity is relative only along the relative motion: events side by side still agree ($\Delta x = 0 \Longrightarrow \Delta t' = 0$). The twin paradox is **not** a violation — that worldline is not straight.




no timelike worldline leaves the 45° cone — that is why no mass reaches c



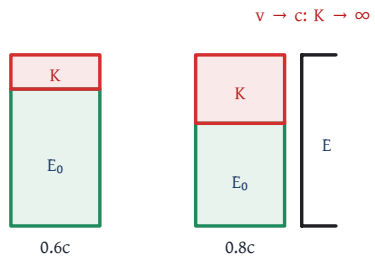

the twin paradox: the straight worldline carries the longest proper time

**IT link:** Two datacentres call different events "simultaneous" — exactly the problem above. Light cones become wall diagrams, latency budgets become $\Delta t$, and the speed of the medium (fibre, $\approx 2 \times 10^8$ m/s) plays the part of c.

# 9 Relativistic Mass, Energy & Momentum

**Definition:** Total energy of a body of rest mass $m_0$ moving at v: $\mathbf{E = \gamma m_0 c^2}$. Writing the historic **relativistic mass** $m = \gamma m_0$, the same statement is $\mathbf{E = mc^2}$, i.e. $\mathbf{E = mc^2 + K}$: the total energy is the rest energy plus the kinetic energy, and $\mathbf{K = (\gamma - 1)m_0 c^2}$.

| Quantity | Formula | Notes |
|---|---|---|
| Mass: rest $m_0$, relativistic m | $m_0 \cdot \gamma m_0$ | $m_0$ is invariant; $m \to \infty$ as $v \to c$ |
| Rest energy $E_0$ | $\mathbf{m_0 c^2}$ | 1 g → $9 \times 10^{13}$ J |
| Total energy E | $\mathbf{\gamma m_0 c^2}$ | $= mc^2$ with $m = \gamma m_0$ |
| Kinetic energy K | $\mathbf{(\gamma - 1)m_0 c^2}$ | $K/E = 1 - 1/\gamma = 1 - \sqrt{(1-\beta^2)}$ |
| Momentum p | $\gamma m_0 v$ | $= mv$ only if $v \ll c$ |
| Mass–energy relation | $E^2 = p^2c^2 + m_0^2c^4$ | photon: $m_0 = 0 \Longrightarrow E = pc$ |




$E_0$ stays; K grows with γ

**Exam trap:** $\mathbf{E = mc^2}$ is the **total** energy only when m means $\gamma m_0$; with the invariant mass $m_0$ it reads $E = \gamma m_0 c^2$. "Mass increases with speed" is old relativistic-mass language — modern texts keep $m_0$ fixed. Charge and $\mathbf{m_0}$ never change with speed, and K is never $\gamma m_0 c^2$.

| v | γ | $m/m_0$ | $K/(m_0c^2)$ | $E/(m_0c^2)$ | K/E | $L/L_0$ | $\Delta t/\Delta t_0$ |
|---|---|---|---|---|---|---|---|
| 0.6c | 1.25 | 1.25 | 0.25 | 1.25 | 0.20 | 0.800 | 1.25 |
| 0.8c | 5/3 = 1.67 | 1.67 | 0.667 | 1.67 | 0.40 | 0.600 | 1.67 |

**Memory trick:** "**E is the whole bar, K is the extra bit on top.**" At 0.6c a quarter of the rest energy has been added (20 % of the total); at 0.8c two thirds has (40 % of the total). Never write $K = \gamma mc^2$.

**Worked example 1 ✍ electron at 0.8c:** $m_0 = 9.11 \times 10^{-31}$ kg, $m_0c^2 = 0.511$ MeV, $\gamma = 5/3$. E = **0.852 MeV**, K = **0.341 MeV**, $m = \mathbf{1.52 \times 10^{-30}}$ **kg**, $p = \mathbf{3.64 \times 10^{-22}}$ **kg·m/s**

**Worked example 2 ✍ proton at 0.6c:** $m_0 = 1.673 \times 10^{-27}$ kg, $m_0c^2 = 938.3$ MeV, $\gamma = 1.25$. E = **1173 MeV** = $1.88 \times 10^{-10}$ J, K = **234.6 MeV**, $p = \gamma m_0 v = \mathbf{3.76 \times 10^{-19}}$ **kg·m/s**

**Worked example 3 ✍ mass–energy in bulk:** 1 g fully released: $E = 10^{-3} \times (3 \times 10^8)^2 = \mathbf{9 \times 10^{13}}$ **J** ≈ 21 500 t of TNT, about 1.4 times the 1945 Hiroshima yield.

**Applications:** Nuclear reactors and weapons · PET imaging ($e^- + e^+ \rightarrow$ two 511 keV γ rays) · smoke detectors ($^{241}$Am α) · the LHC: two 7 TeV protons meet at 14 TeV · γ-ray astronomy, $p = E/c$.

# Night-Before-the-Exam Sheet

**Fact sheet:**

$c = 3 \times 10^8$ m/s (exactly 299 792 458 m/s)

$\gamma = 1/\sqrt{(1 - v^2/c^2)} \geq 1$, $\beta = v/c$

$\Delta t = \gamma \Delta t_0$ ($\Delta t_0$ = proper time)

$L = L_0/\gamma$, along the motion only

$u = (u' + v)/(1 + u'v/c^2)$

$\Delta t' = -\gamma v \Delta x/c^2$ if $\Delta t = 0$

$E = \gamma m_0 c^2$, $E_0 = m_0 c^2$

$K = (\gamma - 1)m_0 c^2$

$m = \gamma m_0$ (historic), $p = \gamma m_0 v$

$E^2 = p^2c^2 + m_0^2c^4$, photon $E = pc$

$\gamma(0.6c) = 1.25$, $\gamma(0.8c) = 1.667$

$\gamma(0.866c) = 2$, $\gamma(0.99c) = 7.09$

1 g $\rightarrow 9 \times 10^{13}$ J

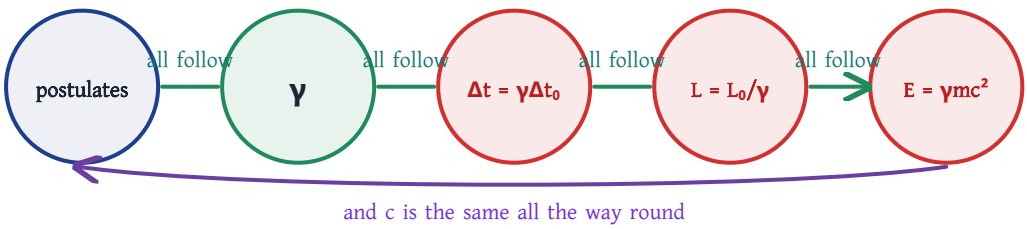
1 eV $= 1.602 \times 10^{-19}$ J

## Top 12 one-liners for MCQs

1. Special relativity: Einstein, **1905**
2. Michelson–Morley ether test: **1887**
3. Postulate 2: c is the same for all inertial frames
4. $\gamma \geq 1$; $\gamma = 1$ only at $v = 0$
5. Proper time is the **shortest** interval
6. Contraction is only along the motion
7. Nothing with mass reaches c
8. Simultaneity is relative along the motion
9. Timelike events keep their order in every frame
10. $K = (\gamma - 1)m_0c^2$, never $\gamma m_0c^2$
11. Photon: $m_0 = 0$ and $E = pc$
12. Michelson: Nobel Prize in Physics, **1907**

**Top traps:** $\gamma(0.6c) = 1.25$, not 1.67 · $\sqrt{(1 - v^2/c^2)}$ is $1/\gamma$, never γ · L contracts along the motion, never across it · $0.6c + 0.6c = 0.882c$, never $1.2c$ · $K \neq \gamma mc^2$ · a null result removes a hypothesis, it does not establish one · c is the speed **in vacuum** only.

**All tricks together:** "The physics is the same, light is the same" (postulates) · γ multiplies time, divides length, grows the energy bar · $\sqrt{(1 - \beta^2)}$ is the length fraction and $1/\gamma$ · $\Delta t = 0$ in the transform makes simultaneity split · proper time is shortest · E is the whole bar, K is the extra bit on top · only side-on lengths shrink.




two postulates in, five results out — every one of them is just γ

## ✓ Practice MCQs (exam pattern, 10 marks)

**1.** Speed of light in vacuum, for all frames:
(a) set by the source (b) larger when closing
(c) same for all frames (d) varies with the frame

**2.** The Lorentz factor at v = 0.8c is:
(a) 5/3 = 1.67 (b) 1.25
(c) 0.60 (d) 1/3

**3.** γ = 2 corresponds to a speed of:
(a) 0.50c (b) 0.99c
(c) 0.866c (d) 0.75c

**4.** Proper length 400 m at 0.8c. Length seen:
(a) 667 m (b) 240 m
(c) 480 m (d) 400 m

**5.** A light clock ticks 2.00 s apart. At 0.6c:
(a) 1.60 s (b) 2.00 s
(c) 3.20 s (d) 2.50 s

**6.** The energy equivalent of 1 g of mass is:
(a) $9 \times 10^{13}$ J (b) $9 \times 10^{16}$ J
(c) $3 \times 10^{5}$ J (d) $9 \times 10^{10}$ J

**7.** Two ships at 0.6c each, same direction:
(a) 1.20c (b) 0.60c
(c) 0.75c (d) 0.882c

**8.** Einstein published it in:
(a) 1901 (b) 1915
(c) 1905 (d) 1921

**9.** The Michelson–Morley experiment (1887):
(a) shift proving ether (b) no shift at all
(c) light slows in water (d) c varies with speed

**10.** Total energy with factor γ ($E_0 = m_0c^2$):
(a) $(\gamma - 1)E_0$ (b) $E_0$
(c) $E_0/\gamma$ (d) $\gamma E_0$

Answer key → 1-c · 2-a · 3-c · 4-b · 5-d · 6-a · 7-d · 8-c · 9-b · 10-d

**Written question:** State Einstein's two postulates. Using the Lorentz transformation, derive time dilation for a clock moving at v = 0.8c, and find the length of a 200 m rod moving along its length at 0.8c. 10 marks

1. **P1** — the laws of physics are identical in every inertial frame. **P2** — light in vacuum travels at c for every inertial observer, whatever the source or observer does. 3
2. Lorentz transform: $x' = \gamma(x - vt)$, $t' = \gamma(t - vx/c^2)$, $\gamma = 1/\sqrt{(1 - v^2/c^2)}$. 2
3. Time dilation: both ticks are at one place in the clock's frame, so $\Delta x' = 0$ and $\Delta x = -v\Delta t'$; then $\Delta t' = \gamma(\Delta t - v\Delta x/c^2) = \Delta t/\gamma$, i.e. $\mathbf{\Delta t = \gamma \Delta t'}$. 3
4. At 0.8c, $\gamma = 1/\sqrt{0.36} = \mathbf{5/3}$: 3.00 s on the clock measures 5.00 s in the lab. 1
5. Length: read both ends at one instant of the measuring frame, $\Delta t = 0$, so $\Delta x' = \gamma \Delta x = L_0$ and $\mathbf{L = L_0/\gamma = 120\ m}$. 1

## Before the exam ✓

- ✓ Can state both postulates
- ✓ Can compute and invert γ
- ✓ Can derive $\Delta t = \gamma \Delta t_0$
- ✓ Knows when $\Delta t' = -\gamma v \Delta x/c^2$ works